Crash Severity Prediction

By David W Ndungu

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Project Overview

- This project focuses on predicting whether a road crash is minor or major using historical crash data. The dataset includes information such as weather conditions, time of day, road conditions, and the number of vehicles involved. By applying machine learning techniques, the project uncovers patterns that influence crash severity, helping to identify high-risk situations. The ultimate goal is to provide insights that support better road safety planning and more effective allocation of emergency resources.

- This project aims to analyze historical crash data to predict the likelihood of a crash being minor or major, allowing stakeholders to make data-driven safety decisions

Business Understanding

Traffic authorities often struggle to identify which conditions or locations are most likely to result in severe crashes, making it difficult to design targeted interventions and allocate resources efficiently.

This leads to increased risk of serious injuries and fatalities, and less effective use of emergency and preventive resources.

Key Question:
How can we predict crash severity to better target safety measures and emergency response efforts?

Stakeholders:

- **Chicago Department of Transportation (or relevant city/state authority)** — responsible for urban road safety planning
- **Emergency Response Services** — need to allocate ambulances and resources efficiently
- **Policy Makers & Public Safety Organizations** — design interventions to reduce severe crashes

Why This Matters ?

- Understanding the factors that contribute to crash severity is critical for improving road safety. By predicting which crashes are more likely to be severe, authorities can plan interventions, optimize emergency response, and focus safety measures on high-risk conditions. This proactive approach can ultimately reduce the impact of road crashes and save lives.

The Data

The dataset used in this analysis contains records of **64,638 traffic crashes**, capturing a wide range of factors that may influence crash severity.

These factors include:

- **Demographic and temporal information**, such as the hour of the crash, day of the week, and region cluster.
- **Environmental conditions**, including weather severity, temperature, and road conditions.
- **Crash-specific characteristics**, such as the number of vehicles involved, crash type, and contributory causes.

The target variable represents whether a crash was **minor or major**, making this a **binary classification problem**.

Key Insights From the Data

- Crash frequency increases steadily from early morning, peaks during late afternoon rush hours (3–5 PM), and declines into the night, reflecting the strong influence of traffic volume and daily human activity patterns on crash occurrence

- crash occurrences are concentrated in specific geographic hotspots, likely influenced by higher traffic volumes or urban infrastructure.

- Overcast and Partially Cloudy weather are common during crashes, underscoring the importance of considering weather factors in road safety measures.

- During the spring and early summer, road accidents occur with the highest frequency, likely attributable to increased outdoor activities and social events, including higher instances of drinking.

Modelling Approach

- A **Logistic Regression model** was first selected as the baseline because it is simple, easy to interpret, and suitable for predicting two outcomes: minor or major crashes.
- The model was then improved by handling class imbalance and adding **new features** such as interactions between time of day and weather, as well as binned temperature categories, to help the model better capture important patterns.
- Advanced models, including **Random Forest and XGBoost**, were also applied to capture more complex relationships in the data and improve prediction accuracy.
- The models were evaluated using metrics such as recall, precision, and F1-score, with a focus on **r**ecall for severe crashes, ensuring the model identifies high-risk situations accurately.
- This step-by-step, iterative approach allowed for continuous improvement and ensured that the modeling results aligned with the goal of supporting road safety and resource planning.

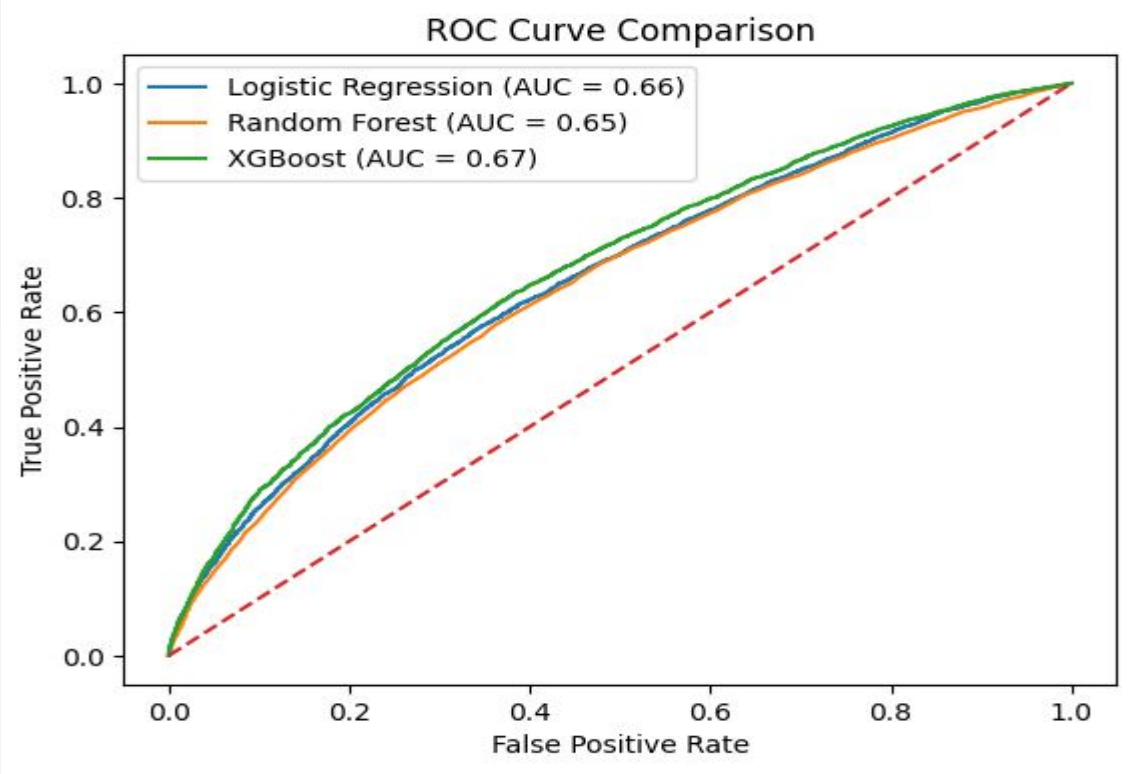
Model Performance

Our best model, **XGBoost**, was able to achieve a balanced performance across both minor and major crashes. It correctly identified approximately 65% of minor crashes and 60% of major crashes, improving detection of high-risk situations compared to earlier models.

The model captures **complex patterns** in the data, including interactions between weather conditions, time of day, and the number of vehicles involved. This allows it to predict which crashes are likely to be more severe, providing actionable insights for road safety planning and emergency response allocation.

While the model is not perfect and may be less accurate for rare or unusual crash scenarios, it offers a **reliable tool for** prioritizing interventions and making data-driven decisions to reduce the impact of severe crashes.

Model Performance Comparison



Recommendations

Stakeholders can use this model to better understand the conditions under which crashes are more likely to be severe. For example:

- Focus safety interventions during high-risk times (e.g., night or adverse weather)
- Improve road conditions in areas associated with higher crash severity
- Allocate emergency response resources more effectively

While the model should not be used as the sole decision-making tool, it provides valuable insights to support data-driven road safety strategies.

Future Work ?

- In the future, the model can be made even better by using real-time traffic and weather information, which would help predict crashes more accurately.
- Adding road location data could show exactly where severe crashes are most likely to happen.

- We could also explore new modeling techniques or extra features to find more patterns. These improvements would make the model more useful for preventing serious crashes and planning road safety.

Thank You For Your Attention !

Presented By:

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Have an **AMAZING** week!